

Summer Term 2026

Investment and Financial Management

Session 07+08: Cost of capital

Berk/DeMarzo (2024): Corporate Finance, 6th edition, Pearson: Chapter 12

-
- 1. The equity cost of capital**
 - 2. The debt cost of capital**
 - 3. A project's cost of capital**
 - 4. Project risk characteristics and financing**
-

1. The equity cost of capital

2. The debt cost of capital

3. A project's cost of capital

4. Project risk characteristics and financing

- **Intuition of the equity cost of capital**

The cost of capital of any investment opportunity equals the expected return of available investments with the same **risk**.

- **Methods of estimating the equity cost of capital**

- The Capital Asset Pricing Model (CAPM)
- Dividend discount model
- Bond yield plus risk premium

■ The Capital Asset Pricing Model (CAPM)

- The CAPM states that the expected return on equity equals the sums of the risk-free interest rate and a premium for bearing the market risk.
- The CAPM estimate / equation is provided by the security market line (SML) equation:

$$r_i = r_f + \beta_i * (E[r_{Mkt}] - r_f)$$

Risk premium for security i (RP_{Mkt})

With:

r_i = Equity cost of capital

r_f = Risk-free rate

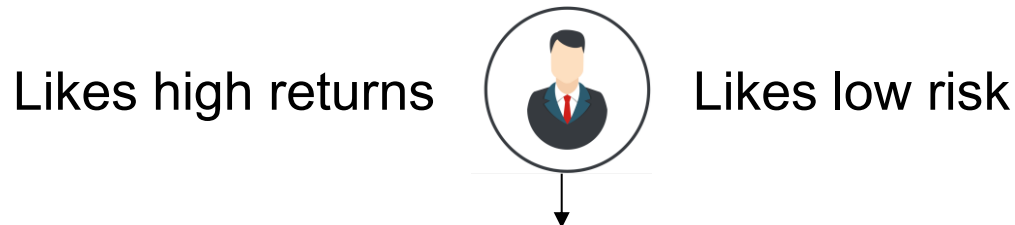
β_i = Return sensitivity of stock i to changes in the market return

$E[r_{Mkt}]$ = Expected return on the market

$E[r_{Mkt}] - r_f$ = Expected market risk premium or **equity risk premium (ERP)**

▪ The Capital Asset Pricing Model (CAPM)

- Two assumptions about the investor:



For a given amount of risk, what's the required return an investor demands as compensation?

- Two types of risk: Only systematic risk is priced in capital markets!
 - Unsystematic (diversifiable risk) vs. undiversifiable (systematic) risk
 - Every investor will hold a portfolio which is fully diversified being the market portfolio to eliminate unsystematic risk components

How to estimate the required return then?

Quantify the amount of systematic risk & determine the risk premium to compensate for this amount of risk!

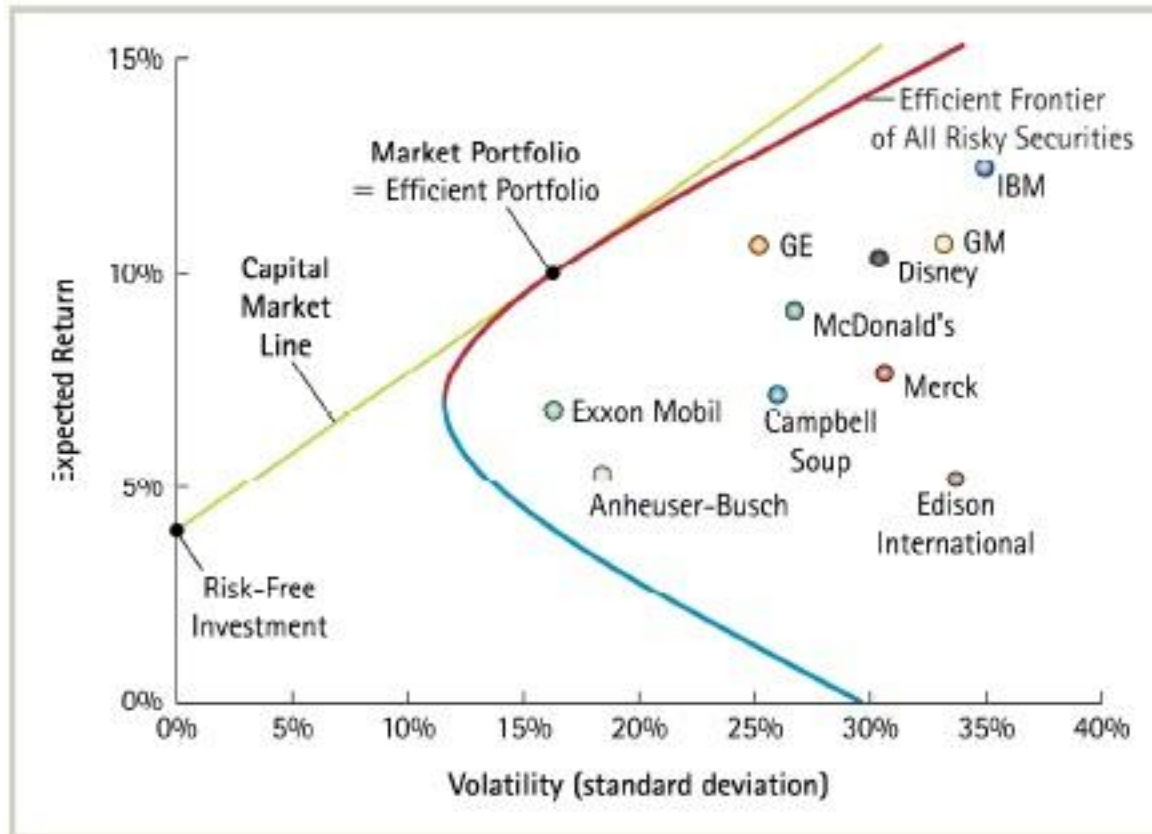
▪ The Capital Asset Pricing Model (CAPM)

Quantify the amount of systematic risk & determine the risk premium to compensate for this amount of risk!

- β_i : To what extent does the security move together with a fully diversified, efficient portfolio that only bears systematic risk (market portfolio)?
- Market risk premium compensates investors for taking on the systematic risk of holding the market portfolio compared to a risk-free investment opportunity (market excess return)
- This leads us to the Security Market Line to estimate the required return for every security or portfolio:

$$r_i = r_f + \beta_i * (E[r_{Mkt}] - r_f)$$

- The Capital Asset Pricing Model (CAPM)



■ The Capital Asset Pricing Model (CAPM)

- The CAPM states that the expected return on equity equals the sums of the risk-free interest rate and a premium for bearing the market risk.
- The CAPM estimate / equation is provided by the security market line (SML) equation:

$$r_i = r_f + \beta_i * (E[r_{Mkt}] - r_f)$$

Risk premium for security i (RP_{Mkt})

With:

r_i = Equity cost of capital

r_f = Risk-free rate

β_i = Return sensitivity of stock i to changes in the market return

$E[r_{Mkt}]$ = Expected return on the market

$E[r_{Mkt}] - r_f$ = Expected market risk premium or **equity risk premium (ERP)**

- **Example of calculating the equity cost of capital**

Computing the Equity Cost of Capital

Problem

Suppose you estimate that Disney's stock (DIS) has a volatility of 20% and a beta of 1.29. A similar process for Domino's Pizza (DPZ) yields a volatility of 30% and a beta of 0.62. Which stock carries more total risk? Which has more market risk? If the risk-free interest rate is 3% and you estimate the market's expected return to be 8%, calculate the equity cost of capital for Disney and Domino's. Which company has a higher cost of equity capital?

▪ Example of calculating the equity cost of capital

Solution

Total risk is measured by volatility; therefore Domino stock has more total risk than Disney. Systematic risk is measured by beta. Disney has a higher beta, so it has more market risk than Domino. Given Disney's estimated beta of 1.29, we expect the price for Disney's stock to move by 1.29% for every 1% move of the market. Therefore, Disney's risk premium will be 1.29 times the risk premium of the market, and Disney's equity cost of capital (from Eq. 12.1) is

$$r_{\text{DIS}} = 3\% + 1.29 \times (8\% - 3\%) = 3\% + 6.45\% = 9.45\%$$

Domino has a lower beta of 0.62. The equity cost of capital for Domino is

$$r_{\text{DPZ}} = 3\% + 0.62 \times (8\% - 3\%) = 3\% + 3.10\% = 6.10\%$$

Because market risk cannot be diversified, it is market risk that determines the cost of capital; thus Disney has a higher cost of equity capital than Domino, even though it is less volatile.

- **The Capital Asset Pricing Model (CAPM)**
 - Steps to implement the CAPM and to calculate the equity cost of capital (r_i):
 - a) Construct the market portfolio and determine its expected excess return ($E[r_{Mkt}] - r_f$) of the (value-weighted) market portfolio
 - b) Estimate the stock's beta β_i = sensitivity to the market portfolio

▪ The Capital Asset Pricing Model (CAPM)

a) Construct the market portfolio and determine its expected excess return ($E[r_{Mkt}] - r_f$) of the (value-weighted) market portfolio

- **Definition of the market portfolio**

It is a value-weighted portfolio of all securities traded in the market. Thus, it is used as a proxy a market index. Market indexes report the value of a particular portfolio of securities.

- **Examples of market indexes:**

- S&P 500: A value-weighted portfolio of the 500 largest U.S. stocks
- Wilshire 5000: A value-weighted index of all U.S. stocks listed on the major stock exchanges
- Dow Jones Industrial Average (DJIA): A price weighted portfolio of 30 large industrial stocks

▪ The Capital Asset Pricing Model (CAPM)

a) Construct the market portfolio and determine its expected excess return ($E[r_{Mkt}] - r_f$) of the (value-weighted) market portfolio

- **Determining the risk-free rate**

- Reflects the yield on U.S. Treasury securities
- Surveys suggest most practitioners use ten-year to 30-year treasuries

- **Estimating the (historical) market risk premium**

- Estimate the risk premium ($E[r_{Mkt}] - r_f$) using the historical average excess return of the market over the risk-free interest rate

S&P 500 Excess Return Versus	Period	
	1926–2015	1965–2015
One-year Treasury	7.7%	5.0%
Ten-year Treasury*	5.9%	3.9%

* Based on a comparison of compounded returns over a ten-year holding period

▪ The Capital Asset Pricing Model (CAPM)

a) Construct the market portfolio and determine its expected excess return ($E[r_{\text{Mkt}}] - r_f$) of the (value-weighted) market portfolio

- **The fundamental market risk premium**

- Two drawbacks of using historical data:
 - Standard errors of the estimates are large
 - Backward looking, so may not represent current expectations
- One alternative is to solve for the discount rate that is consistent with the current level of the index.

$$r_{\text{Mkt}} = \frac{\text{Div}_1}{P_0} + g = \text{Dividend yield} + \text{Expected dividend growth rate}$$

▪ The Capital Asset Pricing Model (CAPM)

b) Estimate the stock's beta β_i = sensitivity to the market portfolio

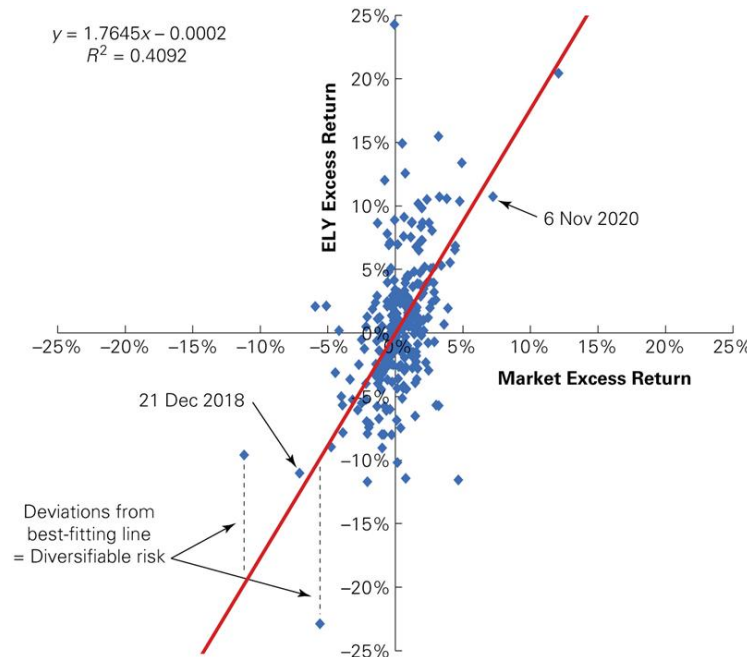
• Estimating beta from historical returns

- Beta measures a security's sensitivity to market risk
- Beta is the expected percent change in the excess return of the security for a 1% change in the excess return of the market portfolio
- **Method 1:** Beta is the slope of the best-fitting line in the plot of a security's excess returns vs. the market's excess returns ➔ Regression analysis
- **Method 2:** Beta is calculated as the covariance of a stock with the market divided by the variance of the market:
 - Stock beta > 1.0: Stock is riskier than the overall market and has higher expected returns
 - Stock beta < 1.0: Stock is less risky than the overall market and is expected to have lower returns

■ The Capital Asset Pricing Model (CAPM)

b) Estimate the stock's beta β_i = sensitivity to the market portfolio

- **Example of estimating beta from historical returns:** Scatterplot of Weekly Excess Returns for Callaway Golf Versus the S&P 500, 2017–2021

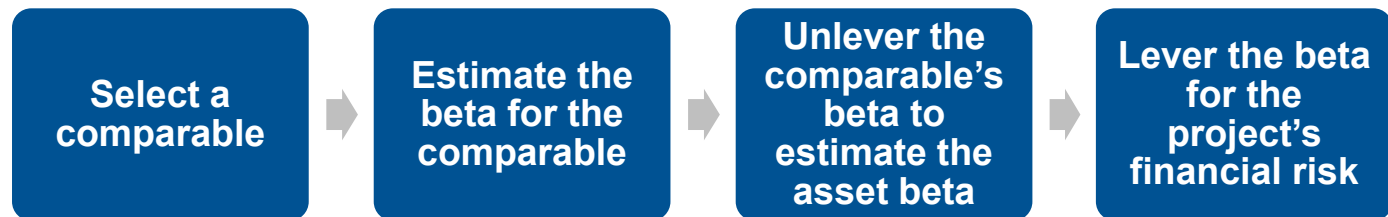


■ The Capital Asset Pricing Model (CAPM)

b) Estimate the stock's beta β_i = sensitivity to the market portfolio

• Issues in estimating beta from historical returns

- Judgement regarding the estimation period, the appropriate market index, the use of a smoothing technique, and adjustments for non-public or small company stocks.
- **For non-public companies**, we need to use comparable companies (with similar risk, size, etc.) to estimate beta.



$$\beta_{\text{Asset}} = \beta_{\text{Equity}} \times \left[\frac{1}{1 + \left((1 - \tau) \times \frac{D}{E} \right)} \right]$$

$$\beta_{\text{Equity}} = \beta_{\text{Asset}} \times \left[1 + \left((1 - \tau) \times \frac{D}{E} \right) \right]$$

■ The Capital Asset Pricing Model (CAPM)

b) Estimate the stock's beta β_i = sensitivity to the market portfolio for non-public companies

Select a comparable



Estimate the beta for the comparable

$$\beta_{\text{Equity}}^C$$



Unlever the comparable's beta to estimate the asset beta

$$\beta_{\text{Asset}}^C = \beta_{\text{Equity}}^C \times \left[\frac{1}{1 + \left((1 - \tau) \times \frac{D^C}{E} \right)} \right]$$



Lever the beta for the project's financial risk

$$\beta_{\text{Equity}}^P = \beta_{\text{Asset}}^C \times \left[1 + \left((1 - \tau) \times \frac{D^P}{E} \right) \right]$$

- **Regression estimate to the Capital Asset Pricing Model (CAPM)**

Using Regression Estimates to Estimate the Equity Cost of Capital Problem

Suppose the risk-free interest rate is 3%, and the market risk premium is 5%. What range for Callaway's equity cost of capital is consistent with the 95% confidence interval for its beta?

▪ Regression estimate to the Capital Asset Pricing Model (CAPM)

Solution

Using the data from 2017 to 2021, and applying the CAPM equation, the estimated beta of 1.76 implies an equity cost of capital of $3\% + 1.76 \times 5\% = 11.8\%$ for Callaway. But our estimate is uncertain, and the 95% confidence interval for Callaway's beta of 1.5 to 2.0 gives a range for Callaway's equity cost of capital from $3\% + 1.5 \times 5\% = 10.5\%$ to $3\% + 2.0 \times 5\% = 13\%$.

-
1. The equity cost of capital
 2. The debt cost of capital
 3. A project's cost of capital
 4. Project risk characteristics and financing
-

▪ **Definition of the debt cost of capital**

The debt cost of capital is relatively easy to calculate and should reflect the current (debt) market rate the company is paying.

▪ **Methods of calculating the debt cost of capital**

• **Yield-to-maturity approach**

- Debt cost of capital = Yield to maturity on a firm's outstanding debt

• **Debt-rating approach**

- Debt cost of capital = Yield based on similarly rated debt (bonds) with similar maturity

• **Beta-CAPM approach**

- Debt cost of capital = Expected return for debt based on its beta using the CAPM

- **Yield-to-maturity approach**

- **Definition of yield-to-maturity**

Yield to maturity (YTM) is the IRR an investor will earn from holding the bond to maturity and receiving its promised payments.

- **The accuracy of the estimate depends on the risk of firm default!**

- **Low risk of firm default:** YTM is a reasonable estimate of investors' expected rate of return.
- **High risk of firm default:** YTM will overstate investors' expected return.

▪ Yield-to-maturity approach

Example: Relation between YTM and risk (I)

- Consider a one-year bond with YTM of y
- For each \$1 invested in the bond today, the issuer promises to pay $\$(1 + y)$ in one year
- Suppose the bond will default with probability p , in which case bond holders receive only $\$(1 + y - L)$, where L is the expected loss per \$1 of debt in the event of default
- The expected return of the bond is:

$$\begin{aligned} r_d &= (1 - p)y + p(y - L) = y - pL \\ &= \text{Yield to maturity} - \text{Prob}(\text{default}) \times \text{Expected loss rate} \end{aligned}$$

➡ The importance of the adjustment depends on the riskiness of the bond!

- Yield-to-maturity approach

Example: Relation between YTM and risk (II)

- Annual default rates by debt rating (1983–2011)

Rating:	AAA	AA	A	BBB	BB	B	CCC	CC-C
Default Rate:								
Average	0.0%	0.1%	0.2%	0.5%	2.2%	5.5%	12.2%	14.1%
In Recessions	0.0%	1.0%	3.0%	3.0%	8.0%	16.0%	48.0%	79.0%

Source: “Corporate Defaults and Recovery Rates, 1920–2011,” *Moody’s Global Credit Policy*, February 2012.

- The average loss rate for unsecured debt is 60%. According to Table 12.2, during average times the annual default rate for B-rated bonds is 5.5%.
- So, the expected return to B-rated bondholders during average times is $0.055 \times 0.60 = 3.3\%$ below the bond’s quoted yield.

■ Beta-CAPM approach

- Alternatively, we can estimate the debt cost of capital using the CAPM.
- Debt betas are difficult to estimate because corporate bonds are traded infrequently.
- One approximation is to use estimates of betas of bond indices by rating category.

By Rating	<i>A and above</i>	<i>BBB</i>	<i>BB</i>	<i>B</i>	<i>CCC</i>
Avg. Beta	< 0.05	0.10	0.17	0.26	0.31
By Maturity	(BBB and above)	<i>1–5 Year</i>	<i>5–10 Year</i>	<i>10–15 Year</i>	<i>> 15 Year</i>
Avg. Beta		0.01	0.06	0.07	0.14
<i>Source: S. Schaefer and I. Strebulaev, “Risk in Capital Structure Arbitrage,” Stanford GSB working paper, 2009.</i>					

- **Yield-to-maturity approach and beta-CAPM approach**

Estimating the Debt Cost of Capital

Problem

In mid-2012, homebuilder KB Home had outstanding 6-year bonds with a yield to maturity of 6% and a B rating. If corresponding risk-free rates were 1%, and the market risk premium is 5%, estimate the expected return of KB Home's debt.

- **Yield-to-maturity approach and beta-CAPM approach**

Solution

Given the low rating of debt, we know the yield to maturity of KB Home's debt is likely to significantly overstate its expected return. Using the average estimates in Table 12.2 and an expected loss rate of 60%, from Eq. 12.7 we have

$$r_d = 6\% - 5.5\%(0.60) = 2.7\%$$

Alternatively, we can estimate the bond's expected return using the CAPM and an estimated beta of 0.26 from Table 12.3. In that case,

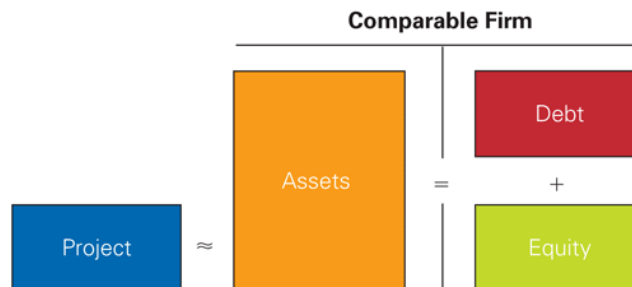
$$r_d = 1\% + 0.26(5\%) = 2.3\%$$

While both estimates are rough approximations, they both confirm that the expected return of KB Home's debt is well below its promised yield.

-
1. The equity cost of capital
 2. The debt cost of capital
 3. A project's cost of capital
 4. Project risk characteristics and financing
-

3. A project's cost of capital

- **Recall our basic question in corporate finance: Should we undertake a given project?**
 - **Assumptions:**
 - All-equity financing of the project separate from any financing decision
 - Ignoring financing risk
 - **Answer: It depends, esp. on the project's cost of capital!**
 - To calculate the project's cost of capital, we need to find a comparable company and estimate the cost of capital of assets as a proxy for the project's cost of capital.
 - All-equity firm comparables
 - Levered firm comparables



- **All-equity comparables**

Estimating the Beta of a Project from a Single-Product Firm

Problem

You have just graduated with an MBA, and decide to pursue your dream of starting a line of designer clothes and accessories. You are working on your business plan, and believe your firm will face similar market risk to Lululemon (LULU).

To develop your financial plan, estimate the cost of capital of this opportunity assuming a risk-free rate of 3% and a market risk premium of 5%.

- **All-equity comparables**

Solution

Checking Yahoo! Finance, you find that Lululemon has no debt. Using five years of weekly data, you estimate their beta to be 1.2. Using LULU's beta as the estimate of the project beta, we can apply Eq. 12.1 to estimate the cost of capital of this investment opportunity as

$$r_{\text{project}} = r_f + B_{\text{LULU}} (E[R_{\text{Mkt}}] - r) = 3\% + 1.20 \times 5\% = 9\%$$

Thus, assuming your business has a similar sensitivity to market risk as Lululemon, you can estimate the appropriate cost of capital as 9%. In other words, rather than investing in the new business, you could invest in the fashion industry simply by buying LULU stock. Given this alternative, to be attractive, the new investment must have an expected return at least equal to that of LULU, which from the CAPM is 9%.

- **Levered firms as comparables: Asset cost of capital / unlevered cost of capital**
 - Expected return required by investors to hold the firm's underlying assets
 - Weighted average of the firm's equity and debt costs of capital:

$$r_U = r_E \frac{E}{E + D} + r_D \frac{D}{E + D}$$

- **All-equity financing: Asset (unlevered) beta**

$$\beta_U = \beta_E \frac{E}{E + D} + \beta_D \frac{D}{E + D}$$

- **Levered firms as comparables: Asset (unlevered) beta**

Unlevering the Cost of Capital

Problem

Your firm is considering expanding its household products division. You identify Procter & Gamble (PG) as a firm with comparable investments. Suppose PG's equity has a market capitalization of \$144 billion and a beta of 0.57. PG also has \$37 billion of AA-rated debt outstanding, with an average yield of 3.1%. Estimate the cost of capital of your firm's investment given a risk-free rate of 3% and a market risk-premium of 5%.

- **Levered firms as comparables: Asset (unlevered) beta**

Solution

Because investing in this division is like investing in PG's assets by holding its debt and equity, we can estimate our cost of capital based on PG's unlevered cost of capital. First, we estimate PG's equity cost of capital using the CAPM as $r_E = 3\% + 0.57(5\%) = 5.85\%$. Because PG's debt is highly rated, we approximate its debt cost of capital using the debt yield of 3.1%. Thus, PG's unlevered cost of capital is

$$r_U = \frac{144}{144 + 37} 5.85\% + \frac{37}{144 + 37} 3.1\% = 5.29\%$$

Alternatively, we can estimate PG's unlevered beta. Given its high rating, if we assume PG's debt beta is zero we have

$$\beta_U = \frac{144}{144 + 37} 0.57 + \frac{37}{144 + 37} 0 = 0.453$$

Taking this result as an estimate of the beta of our project, we can compute our project's cost of capital from the CAPM as $r_U = 3\% + 0.453(5\%) = 5.27\%$.

The slight difference in r_U using the two methods arises because in the first case, we assumed the expected return of PG's debt is equal to its promised yield of 3.1%, while in the second case, we assumed the debt has a beta of zero, which implies an expected return equal to the risk-free rate of 3% according to the CAPM. The truth is somewhere between the two results, as PG's debt is not completely risk-free.

▪ Cash and net debt

- Some firms maintain high cash balances
- Cash is a risk-free asset that reduces the average risk of the firm's assets
- Since the risk of the firm's enterprise value is what we are concerned with, leverage should be measured in terms of net debt in our calculations:

$$D = \text{Net debt} = \text{Debt} - \text{Excess cash and short-term investments}$$

- **Cash and net debt**

Problem

In mid-2022, Garmin Ltd. had a market capitalization of \$18.8 billion, \$100 million in debt, and \$1.6 billion in cash. If its estimated equity beta was 0.93, estimate the beta of Garmin's underlying business enterprise.

- **Cash and net debt**

Solution

Garmin has net debt = $(0.1 - 1.6) = -\$1.5$ billion, and thus has an enterprise value of $(18.8 - 1.5) = \$17.3$ billion.

Assuming Garmin's debt and cash investments are both risk-free, we can estimate the beta of this enterprise value as:

$$B_U = \frac{E}{E + D} B_E + \frac{D}{E + D} B_D = \frac{18.8}{18.8 - 1.5} 0.93 + \frac{-1.5}{18.8 - 1.5} 0 = 1.01$$

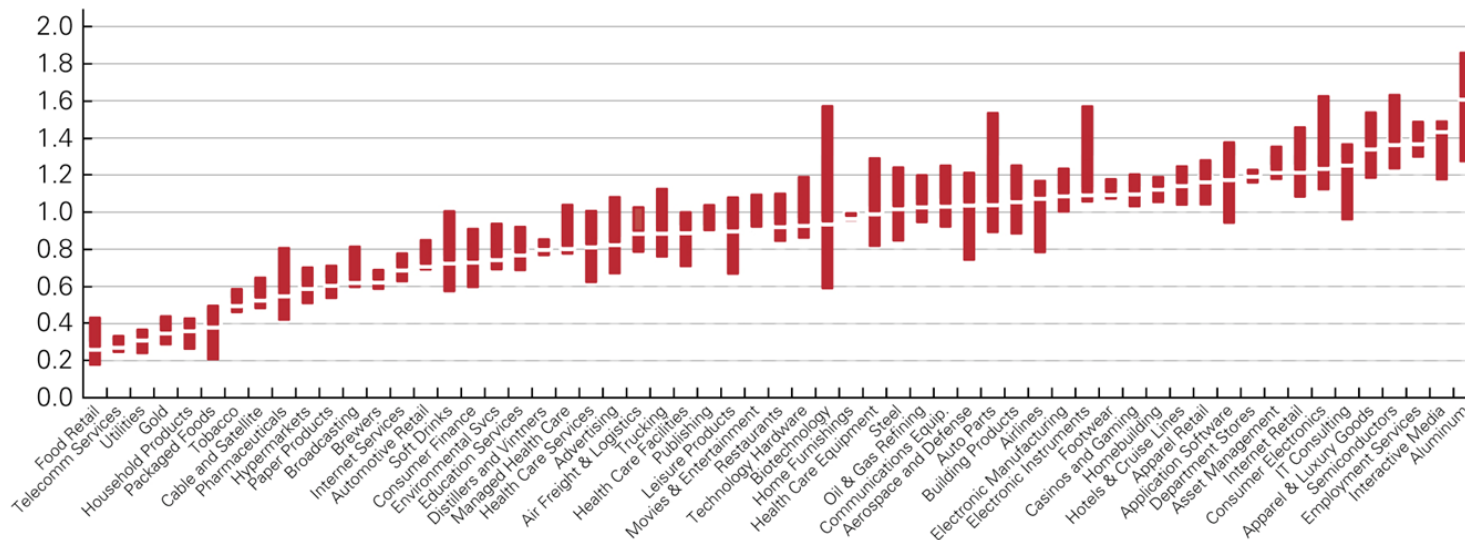
Note that in this case, Garmin's equity is *less* risky than its underlying business activities due to its cash holdings.

3. A project's cost of capital

■ Industry asset beta

- After adjusting for the leverage of different firms to determine their asset betas, we can combine estimates of asset betas for multiple firms in the same industry.
- Doing this will reduce the estimation error of the estimated beta for the project.

Asset Beta



Source: Author calculations based on data from Capital IQ.

- Industry asset beta

Estimating an Industry Asset Beta

Problem

Consider the following data for U.S. department stores in mid-2009, showing the equity beta, ratio of net debt to enterprise value (D/V), and debt rating for each firm. Estimate the average and median asset beta for the industry.

Company	Ticker	Equity Beta	D/V	Debt Rating
Dillard's	DDS	2.38	0.59	B
J. C. Penney Company	JCP	1.60	0.17	BB
Kohl's	KSS	1.37	0.08	BBB
Macy's	M	2.16	0.62	BB
Nordstrom	JWN	1.94	0.35	BBB
Saks	SKS	1.85	0.50	CCC
Sears Holdings	SHLD	1.36	0.23	BB

3. A project's cost of capital

■ Industry asset beta

Solution

Note that D/V provides the fraction of debt financing, and $(1 - D/V)$ the fraction of equity financing, for each firm. Using the data for debt betas from Table 12.3, we can apply Eq. 12.9 for each firm. For example, for Dillard's:

$$\beta_U = \frac{E}{E+D}\beta_E + \frac{D}{E+D}\beta_D = (1 - 0.59)2.38 + (0.59)0.26 = 1.13$$

Doing this calculation for each firm, we obtain the following estimates:

Ticker	Equity Beta	D/V	Debt Rating	Debt Beta	Asset Beta
DDS	2.38	0.59	B	0.26	1.13
JCP	1.60	0.17	BB	0.17	1.36
KSS	1.37	0.08	BBB	0.10	1.27
M	2.16	0.62	BB	0.17	0.93
JWN	1.94	0.35	BBB	0.10	1.30
SKS	1.85	0.50	CCC	0.31	1.08
SHLD	1.36	0.23	BB	0.17	1.09
				Average	1.16
				Median	1.13

The large differences in the firms' equity betas are mainly due to differences in leverage. The firms' asset betas are much more similar, suggesting that the underlying businesses in this industry have similar market risk. By combining estimates from several closely related firms in this way, we can get a more accurate estimate of the beta for investments in this industry.

-
1. The equity cost of capital
 2. The debt cost of capital
 3. A project's cost of capital
 4. Project risk characteristics and financing
-

■ Differences in project risk

- Firm asset betas reflect market risk of the average project in a firm
- Individual projects may be more or less sensitive to market risk
- Financial managers in multi-divisional firms should evaluate projects based on asset betas of firms in a similar line of business
- Another factor that can affect market risk of a project is its degree of operating leverage:
 - Operating leverage is the relative proportion of fixed versus variable costs
 - A higher proportion of fixed costs increases the sensitivity of the project's cash flows to market risk:
 - The project's beta will be higher
 - A higher cost of capital should be assigned

- **Operating leverage and beta**

Operating Leverage and Beta

Problem

Consider a project with expected annual revenues of \$120 and costs of \$50 in perpetuity. The costs are completely variable, so that the profit margin of the project will remain constant. Suppose the project has a beta of 1.0, the risk-free rate is 5%, and the expected return of the market is 10%. What is the value of this project? What would its value and beta be if the revenues continued to vary with a beta of 1.0, but the costs were instead completely fixed at \$50 per year?

■ Operating leverage and beta

Solution

The expected cash flow of the project is $\$120 - \$50 = \$70$ per year. Given a beta of 1.0, the appropriate cost of capital is $r = 5\% + 1.0(10\% - 5\%) = 10\%$. Thus, the value of the project if the costs are completely variable is $\$70/10\% = \700 .

If instead the costs are fixed, then we can compute the value of the project by discounting the revenues and costs separately. The revenues still have a beta of 1.0, and thus a cost of capital of 10%, for a present value of $\$120/10\% = \1200 . Because the costs are fixed, we should discount them at the risk-free rate of 5%, so their present value is $\$50/5\% = \1000 . Thus, with fixed costs the project has a value of only $\$1200 - \$1000 = \$200$.

What is the beta of the project now? We can think of the project as a portfolio that is long the revenues and short the costs. The project's beta is the weighted average of the revenue and cost betas, or

$$\beta_P = \frac{R}{R - C} \beta_R - \frac{C}{R - C} \beta_C = \frac{1200}{1200 - 1000} 1.0 - \frac{1000}{1200 - 1000} 0 = 6.0$$

Given a beta of 6.0, the project's cost of capital with fixed costs is $r = 5\% + 6.0(10\% - 5\%) = 35\%$. To verify this result, note that the present value of the expected profits is then $\$70/35\% = 200$. As this example shows, increasing the proportion of fixed versus variable costs can significantly increase a project's beta (and reduce its value).

- **The weighted average cost of capital (WACC)**

- **Perfect capital markets**

In perfect capital markets, the choice of financing does not affect cost of capital or project NPV

- **Taxes – A big imperfection**

When interest payments on debt are tax deductible, the net cost to the firm is given by:

$$\text{Effective after-tax interest rate} = r \times (1 - \tau_c)$$

- **The weighted average cost of capital (WACC)**

$$r_{WACC} = r_E \frac{E}{E + D} + r_D \frac{D}{E + D} (1 - T_c)$$

- **Given a target leverage ratio:**

$$r_{WACC} = r_U - \frac{D}{E + D} T_c r_D$$

▪ r_{WACC} vs. r_U

- Unlevered cost of capital (or pre-tax WACC) = r_U
 - Expected return investors will earn by holding the firm's assets
 - In a world with taxes, it can be used to evaluate an all-equity project with the same risk as the firm
- In a world with taxes, WACC is less than the expected return of the firm's assets
 - With taxes, WACC can be used to evaluate a project with the same risk and the same financing as the firm

- **Calculation of WACC**

Estimating the WACC

Problem

Dunlap Corp. has a market capitalization of \$100 million, and \$25 million in outstanding debt. Dunlap's equity cost of capital is 10%, and its debt cost of capital is 6%. What is Dunlap's unlevered cost of capital? If its corporate tax rate is 40%, what is Dunlap's weighted average cost of capital?

■ Calculation of WACC

Solution

Dunlap's unlevered cost of capital, or pretax WACC, is given by

$$r_U = \frac{E}{E + D} r_E + \frac{D}{E + D} r_D = \frac{100}{125} 10\% + \frac{25}{125} 6\% = 9.2\%$$

Thus, we would use a cost of capital of 9.2% to evaluate all-equity financed projects with the same risk as Dunlap's assets.

Dunlap's weighted average cost of capital, or WACC, can be calculated using either Eq. 12.12 or 12.13:

$$\begin{aligned} r_{wacc} &= \frac{E}{E + D} r_E + \frac{D}{E + D} r_D (1 - \tau_C) = \frac{100}{125} 10\% + \frac{25}{125} 6\% (1 - 40\%) = 8.72\% \\ &= r_U - \frac{D}{E + D} \tau_C r_D = 9.2\% - \frac{25}{125} (40\%) 6\% = 8.72\% \end{aligned}$$

We can use the WACC of 8.72% to evaluate projects with the same risk and the same mix of debt and equity financing as Dunlap's assets. It is a lower rate than the unlevered cost of capital to reflect the tax deductibility of interest expenses.

■ Chapter quiz of sessions 07+08

1. What inputs do we need to estimate a firm's equity cost of capital using the CAPM?
2. How do you determine the weight of a stock in the market portfolio?
3. How can you estimate the market risk premium?
4. How can you estimate a stock's beta from historical returns?
5. Why does the yield to maturity of a firm's debt generally overestimate its debt cost of capital?
6. Describe two methods that can be used to estimate a firm's debt cost of capital.
7. What data can we use to estimate the beta of a project?
8. Even if the CAPM is not perfect, why might we continue to use it in corporate finance?